Huawei Access Node HEB-ACC-33 Routing Protocol Flap Resolution SOP (Hebbal Edge)

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3060

Description

The Routing Protocol Flap failure domain on Huawei Access Node HEB-ACC-33 (Role: Access) at Hebbal Edge Site Location is primarily caused by a cascading effect of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This is often triggered by a combination of factors including, but not limited to, misconfiguration, software bugs, or hardware issues within the node's routing protocol modules. The Huawei Access Node, being a key component in the Hebbal Edge network, is responsible for facilitating high-speed data transmission between various network segments. When the routing protocol modules on this node malfunction, it leads to a series of events that result in a routing protocol flap, causing network instability and packet loss. It is essential to identify the root cause of this failure and perform necessary corrective actions to prevent future occurrences.

Pre-requisite Checks

- item: Verify Power and Connectivity, description: Ensure power is on and all cables are securely connected to the Huawei Access Node HEB-ACC-33.

- item: Check Redundant Links, description: Verify the status of redundant links to ensure they are functioning correctly.

- item: Confirm Network Isolation, description: Isolate the affected network segment to prevent further impact on the overall network.

Potential Root Causes

Cause	Probability
Routing Protocol Misconfiguration	High
Software Bug or Hardware Issue	Medium
Environmental Factors (Temperature, Humidity)	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Display BGP routing table information., Description: This command helps identify any BGP route churn issues by displaying the BGP routing table information.

Command: display ospf neighbor, Expected Output: Display OSPF neighbor information., Description: This command helps identify any OSPF adjacency loss issues by displaying the OSPF neighbor information.
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```
Command: display isis lsp flooding, Expected Output: Display ISIS LSP flooding information., Description: This command helps identify any ISIS LSP flooding issues by displaying the ISIS LSP flooding information.
```

Resolution Steps

Step 1: Step: Reset BGP Session, Command: reset bgp session, Description: Reset the BGP session to clear any existing routing information and re-establish a new session.

Step 2: Step: Re-configure Routing Protocol Parameters, Command: system-view router bgp bgp router-id XXX.XXX.XXX.XXX bgp confederation id XXX bgp default local-preference XXX exit commit, Description: Re-configure the routing protocol parameters to ensure correct settings are applied.

Step 3: Step: Verify Routing Protocol Configuration, Command: display bgp confederation, Description: Verify the routing protocol configuration to ensure it matches the expected settings.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Verify BGP routing table information., Description: Verify the BGP routing table information to ensure it matches the expected output.

Command: display ospf neighbor, Expected Output: Verify OSPF neighbor information., Description: Verify the OSPF neighbor information to ensure it matches the expected output.

Command: display isis lsp flooding, Expected Output: Verify ISIS LSP flooding information., Description: Verify the ISIS LSP flooding information to ensure it matches the expected output.
```

Roll-back Procedures

- Step: Restore Previous Configuration, Command: system-view undo router bgp commit, Description: Restore the previous configuration to revert any changes made during the resolution process.

- Step: Reset BGP Session, Command: reset bgp session, Description: Reset the BGP session to clear any existing routing information and re-establish a new session.